General Mathematics — Exercise Sheet 1

Statements, sets and proofs

Variant: Statements

Exercise 1 (A choice depending on the input). Let $A = \mathbb{R} \setminus \{1\}$. Compare the statements

$$P: \ \forall x \in A \ \exists y \in \mathbb{R}: \ x + y = xy, \qquad Q: \ \exists y \in \mathbb{R} \ \forall x \in A: \ x + y = xy.$$

Write each negation with all quantifiers. Decide whether P and Q are true. For a true statement give a proof. For a false statement prove its negation. Use Lecture 1's quantifier and negation rules.

Exercise 2 (Membership and a false set identity). Let A, B, C be subsets of a set U. Define their symmetric difference by $A \triangle B = (A \setminus B) \cup (B \setminus A)$.

(a) Prove $A \triangle B = (A \cup B) \setminus (A \cap B)$ by membership.

(b) Give a counterexample to $A \setminus (B \cup C) = (A \setminus B) \cup (A \setminus C)$. State and prove the correct formula with an intersection on the right.

Use Lecture 1's definitions of union, intersection and set difference.

Exercise 3 (A contrapositive and its converse). For $x, y \in \mathbb{R}$, prove by contraposition that

$$x + y > 2 \quad \Longrightarrow \quad x > 1 \text{ or } y > 1.$$

Test the converse. Also decide whether the conclusion can be replaced by $x > 1$ and $y > 1$. Give explicit counterexamples wherever a claim fails.

Exercise 4 (Rational and irrational arithmetic). Let $r \in \mathbb{Q}$ and $s \in \mathbb{R} \setminus \mathbb{Q}$. Prove $r + s \notin \mathbb{Q}$. Determine exactly which rational r guarantee $rs \notin \mathbb{Q}$. Finally, give two irrational real numbers whose product is rational. You may use Lecture 1's irrationality of $\sqrt{2}$.